IPA Project Report (Phase I) Implementation of a 64-bit RISC-V CPU

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***Abstract*—This report presents the design and implementation of a 64-bit RISC-V CPU as part of the IPA Project Phase I. We provide a comprehensive overview of the computational architecture, focusing on the datapath design, control unit implementation, and instruction execution pipeline. The work encompasses the development of essential arithmetic components, including adders and multipliers, as well as the integration of memory subsystems and instruction decoding logic. Our implementation demonstrates the practical challenges and solutions in modern processor design, particularly regarding the trade-offs between performance, area, and power consumption. The project serves as a foundational study in digital system design and computer architecture [1] principles.**

I. Introduction and Work

BINARY addition is the most fundamental arithmetic operation in modern digital systems, serving as the core of Arithmetic Logic Units (ALUs), microprocessors, and digital signal processing (DSP) architectures. As the demand for portable and high-performance computing devices increases, the design of adders that balance speed, power consumption, and area constraints has become a critical research objective. While simple topologies like the Ripple Carry Adder (RCA) offer compactness, they suffer from a very slow carry propagation path (which depends on previous carry to be calculated to process the next sum), making them unsuitable for high-speed applications as the number of bits increases. Conversely, adders such as the Carry Look-ahead Adder (CLA) perform carry and sum computations in parallel, though often at the cost of increased complexity and power dissipation

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References

[1] J. L. Hennessy and D. A. Patterson, *Computer architecture: a quantitative approach*. Elsevier, 2011.